fileB

The last example, there is no local instance, you are directly operating between two remote locations.

9.1 Public-Key Authentication

You must have wondered at this point that during the automation procedure how exactly do we plan to enter the passwords without human intervention, of course we can store them as default values in our text script, but then we are storing our sensitive passwords in text files that can be accessed by anyone. To counter this, we use Public-Key Authentication systems.

We begin by generating the key pair

```
$ shh-keygen -t rsa -b 2048 -c 'digit@testkey.com'
```

It will ask you for a passphrase twice. Fancy yourself, as for the location, the default will be .ssh in your home directory.

The two files containing your fingerprint will be `~/.ssh/id_rsa` and `~/.ssh/id_rsa.pub`

As you can see, adjusting the `-b` parameter would adjust the size, you can put `8192` if you are a little concerned. The `-c` option is for comment, so you can distinguish the needed key pair from others.

The file with the extension `*.pub` is your public key, placing it on the machine which is running the ssh server, would then allow you easy access courtesy your private key. Do the following to ensure correct permissions on the server

```
$ mkdir -p ~/.ssh
```

assuming it ain't thereafter

```
$chmod 0700 ~/.ssh
```

bullet proofing the permissions.

```
$ cp your_pub_key_file_location ~/.ssh/authorized_keys
```

```
$chmod 0600 ~/.ssh/authorized_keys
```

for adding more keys, you can keep appending to the file.

So basically all your clients will generate the key pair, and then you can